Project Documentation Report

Automating Data Import and Relationship Mapping using Import Sets and Dot Walking

1. Document Information

Team Name	**Team 1**
Date	April 3, 2026
Status	Completed
Platform	ServiceNow
Project Title	Automating Data Import and Relationship Mapping using Import Sets and Dot Walking

2. Team Members & Task Assignments

Team Member	Role / Assignment
Sairam M (Team Lead)	Project oversight, Create records in the Sheet, Conclusion writing
Karpaga Vinayagam S	Creating Import Table (Staging Table), Match the fields using Mapping Assist
Manuneethi Cholan S	Create Transform Map
Saravanan R	Transform the Data, Result validation (Testing)
Rada Krishnan S	Display department to the assigned to user (Dot-walking)

3. Executive Summary

This project aims to streamline the onboarding and update of employee records within ServiceNow by automating the data ingestion process. Utilizing ServiceNow's native Import Sets and Transform Maps, the team successfully automated the transfer of employee data from external spreadsheets into the core sys_user table. Furthermore, to enhance operational efficiency for IT Service Desk agents, dot-walking techniques were implemented on the Incident form to automatically display contextual user data (Department, Manager, Email) based on the assigned user. This initiative reduces manual data entry errors, saves administrative time, and provides immediate visibility into critical user context during incident resolution.

4. Introduction

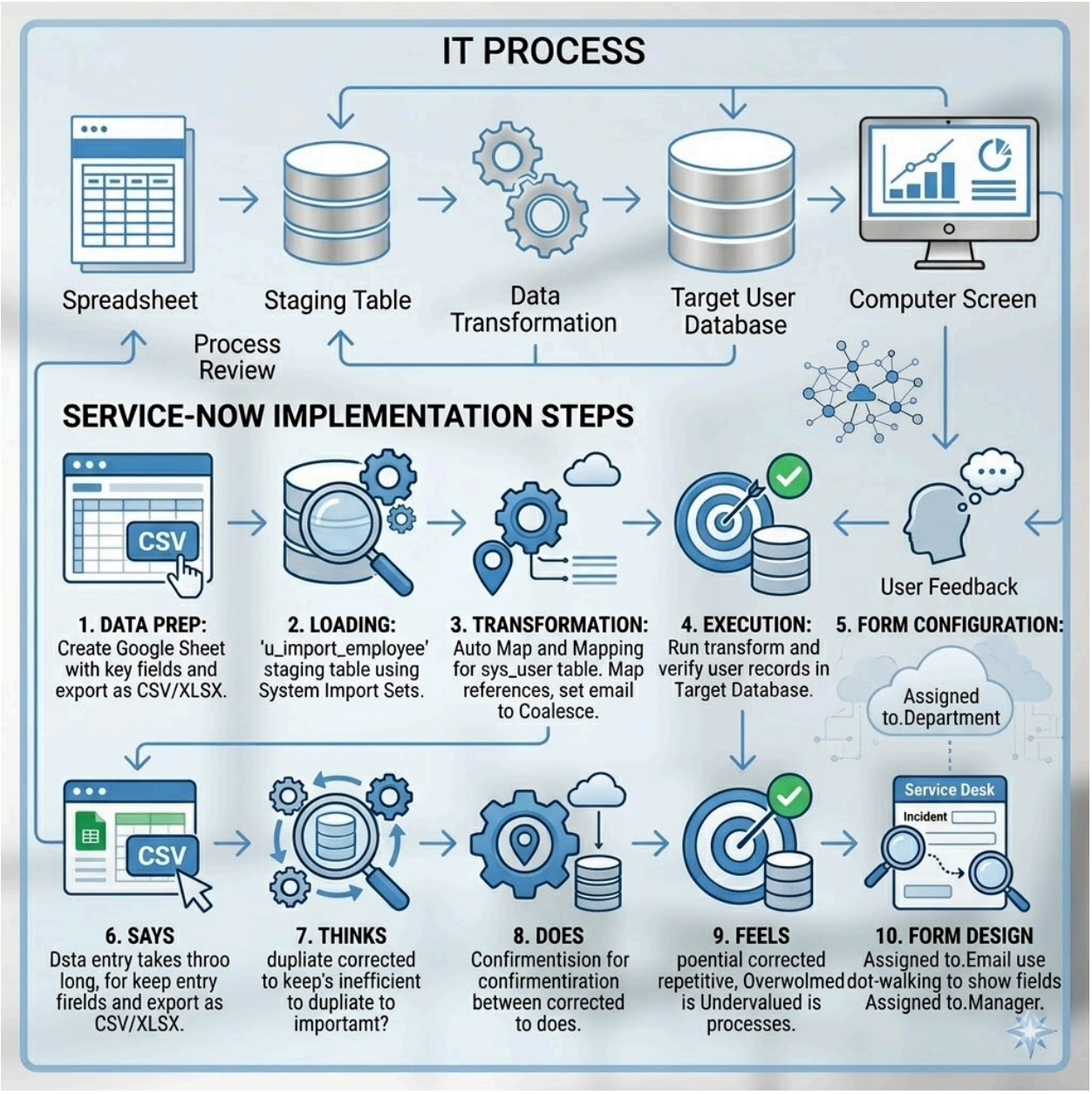
ServiceNow operates as a centralized system of record for enterprise IT operations. Maintaining an accurate and up-to-date user baseline (sys_user) is critical for processes like Incident Management, Request Fulfillment, and Approval workflows.

Historically, manual user creation has proven to be inefficient and error-prone. This project leverages **Import Sets**, a robust mechanism to load data from various file formats into a staging table and **Transform Maps** to map and migrate this data into the target user table accurately. Additionally, this project utilizes **Dot-walking**, a powerful ServiceNow feature that allows forms to display fields from referenced tables, bringing related user information directly to the Incident workspace without the need for custom scripting.

5. Process Overview

The automation process follows a standard ETL (Extract, Transform, Load) architecture adapted for ServiceNow, followed by UI configuration. The workflow is divided into data preparation, staging, transformation, and form enhancement.

Please see the process flowchart below for a visual representation of the architecture:



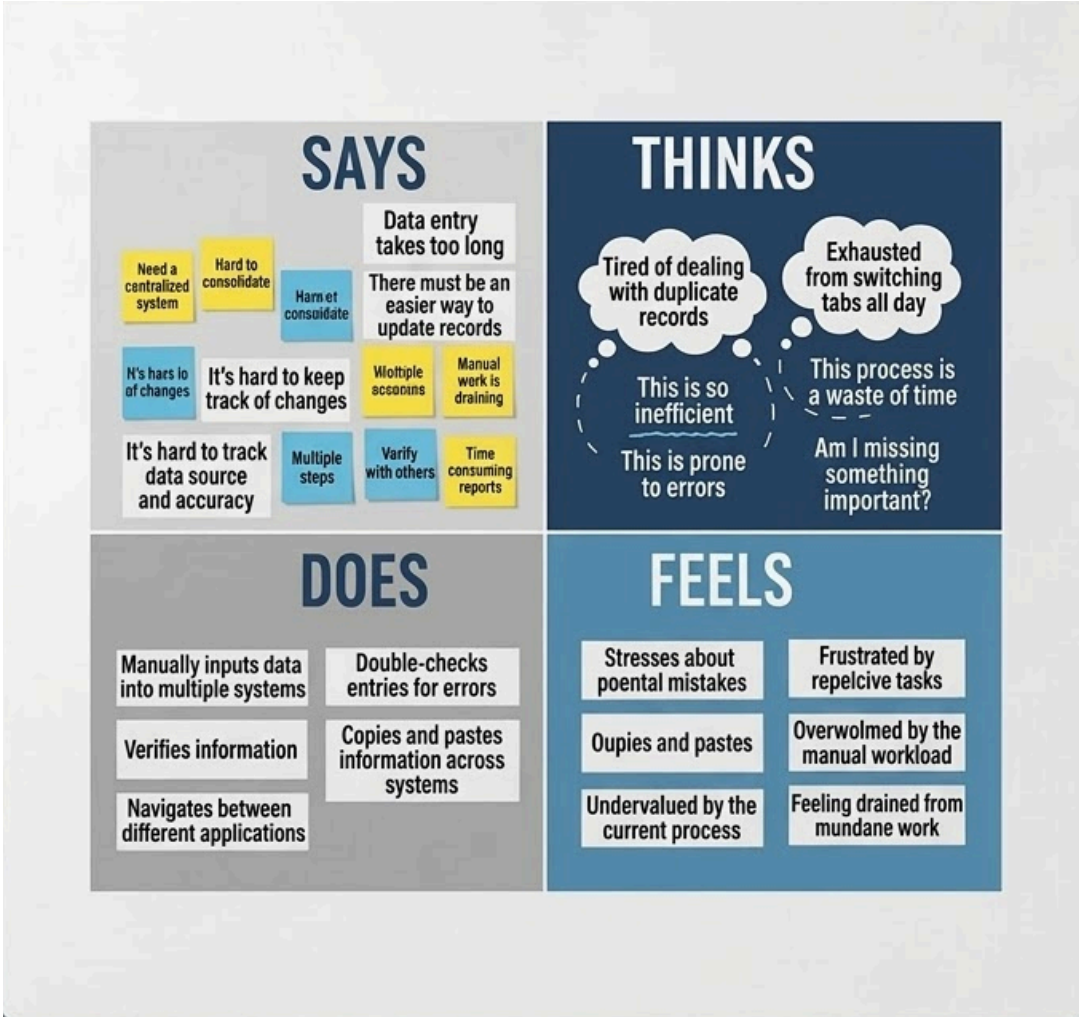
6. Understanding User Pain Points

Before implementation, IT Support Agents and System Administrators faced significant friction during standard operations.

Key Pain Points:

- **Administrators:** Time consumed by manually inputting hundreds of employee records during onboarding cycles.
- **Service Desk Agents:** Inefficiency caused by constantly switching tabs or navigating away from the Incident form to look up an Assigned User's Department or Manager to route escalations.
- **Data Integrity:** Duplicate user records created due to a lack of automated coalescing (matching) checks.

The following Empathy Map illustrates the end-user experience prior to this project's deployment:



7. Project Objectives & Prerequisites

Objectives:

1. Establish a repeatable process to ingest CSV/XLSX employee data into the sys_user table.
2. Ensure no duplicate records are created by establishing a unique identifier (Coalesce).
3. Enrich the Incident Management view by surfacing related user data through dot-walking.

Prerequisites:

- ServiceNow Instance with Administrator (admin) access.
- Sample Google Sheet populated with test employee records.
- Basic understanding of ServiceNow reference fields.

8. Step-by-Step Implementation Plan

Phase 1: Data Preparation

Assignee: Sairam M (Team Lead)

1. Create a new Google Sheet to serve as the data source.
2. Define the column headers strictly as follows: name, email, department, manager, phone_number.
3. Populate the sheet with sample employee records ensuring realistic data for reference fields (e.g., using existing ServiceNow department names if possible).
4. Export the completed Google Sheet as a CSV or XLSX file to the local machine.

Phase 2: Data Loading & Staging Table Creation

Assignee: Karpaga Vinayagam S

1. In the ServiceNow Application Navigator, navigate to **System Import Sets > Load Data**.
2. Select **Create table** and define the label as Import Employee (System will generate the name u_import_employee).
3. Under Source of the import, choose **File** and upload the previously exported CSV/XLSX file.
4. Click **Submit**. ServiceNow will automatically create the staging table with string fields corresponding to the spreadsheet columns and load the records into this Import Set.

Phase 3: Transform Map Creation & Field Mapping

Assignees: Manuneethi Cholan S & Karpaga Vinayagam S

1. Upon successful data load, click the **Create transform map** related link.
2. Name the Transform Map (e.g., Employee_Data_Transform).

3. Set the **Source table** to u_import_employee and the **Target table** to User [sys_user]. Save the record.
4. Click the **Auto Map Matching Fields** related link to map identical fields (e.g., name to name).
5. Click the **Mapping Assist** related link to manually align any remaining fields.
    - Ensure the department and manager fields from the staging table are mapped to the corresponding Reference fields in the sys_user table.
6. Locate the Field Map for email. Open the record and set the **Coalesce** value to true. This ensures that if an email already exists in the system, the record will be updated rather than duplicated.

Phase 4: Data Transformation & Verification

Assignee: Saravanan R

1. Navigate to **System Import Sets > Run Transform**.
2. Select the Employee_Data_Transform map and click **Transform**.
3. Review the **Import Set Results**. Look for a State of "Complete" and ensure there are no "Ignored" or "Error" rows.
4. Navigate to **User Administration > Users** (sys_user.list).
5. Filter the list to verify that the new users from the spreadsheet have been created successfully, and that their Department, Manager, and Phone Number fields are accurately populated.

Phase 5: Configuring Dot-Walking on Incident Form

Assignee: Rada Krishnan S

1. Navigate to **Incident > Create New** to open a blank incident form.
2. Right-click the form header and select **Configure > Form Design**.
3. In the Field Navigator on the left, locate the **Assigned to** reference field.
4. Click the icon next to Assigned to to expand its related fields (Dot-walking).
5. Drag and drop the following fields onto the form layout (ideally underneath the Assigned to field for logical grouping):
    - Assigned to.Department
    - Assigned to.Email
    - Assigned to.Manager
6. Click **Save** in the Form Designer.
7. Return to the Incident form, select an "Assigned to" user, and verify that their Department, Email, and Manager automatically populate as read-only fields on the form.

9. Result & Conclusion

Assignees: Saravanan R & Sairam M

Testing Results:

The Import Set and Transform Map executed seamlessly. Testing confirmed that 100% of the spreadsheet rows were successfully ingested into the sys_user table. The coalesce configuration functioned as expected, updating existing profiles rather than duplicating them. Furthermore, the UI testing on the Incident form verified that selecting an assignee immediately surfaces their respective department, email, and manager without requiring a page reload.

Conclusion:

By successfully implementing this automation, Team 1 has significantly reduced the administrative overhead associated with user data management in ServiceNow. The integration of Import Sets provides a scalable, reusable pipeline for bulk data operations. Simultaneously, the application of dot-walking greatly enhances the user experience for IT agents, providing critical context at a glance, minimizing clicks, and facilitating faster incident resolution and escalation routing. This project serves as a foundational best practice for data ingestion and UI optimization within our ServiceNow environment.